

# Beyond Ingress

Mastering the Kubernetes Gateway API

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Programming With Wolfgang

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# Agenda

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Expose pods

Gateway API

Routing

Gateway API vs. Ingress

Gateway Controller

Demo

# Wolfgang Ofner

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Focus on Azure, Kubernetes, and DevOps



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# Expose Pods

# Expose Pods

Service with type Load Balancer

Ingress

Gateway API

# Ingress and Gateway API Features

- One entry point
- Better routing
- Automate TLS certificates with Cert-Manager
- TLS termination

# Ingress

## Ingress Controller:

- Traefik
- Nginx
- Application Gateway Ingress Controller (AGIC)

# Ingress Downsides

Ingress controllers have a lot of control how traffic is routed

Annotations are unique to the used Ingress controller (e.g. Nginx) → vendor lock-in

No cross-namespace support for TLS secrets



# Ingress Downsides

Ingress does not support traffic policies

No modularity

No real Separation of Concerns

Gateway API GA in 2023 to solve these downside

# Gateway API

# Gateway API

Components are modular and split into multiple parts

Gateway Class configures proxy (similar to Ingress Class)

Gateway Class depends on Gateway Controller

```
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass ←
metadata:
  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

```
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: envoy ←
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

apiVersion: gateway.networking.k8s.io/v1

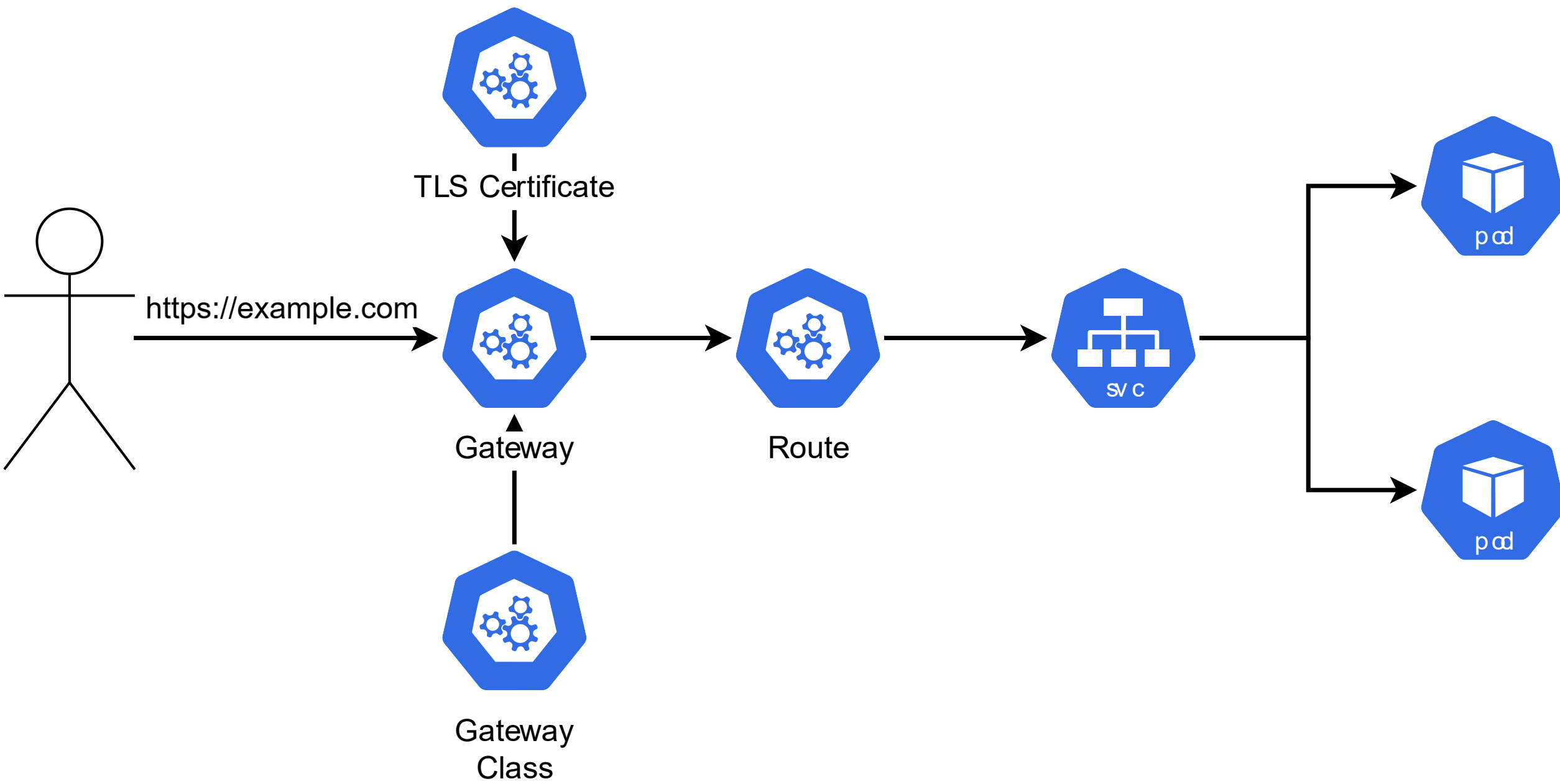
kind: GatewayClass

metadata:

name: envoy

spec:

controllerName: gateway.envoyproxy.io/gatewayclass-controller



# Gateway

Gateway contains listeners

Governs TLS certificates

Namespaced

Gateway consumes global Gateway Class



apiVersion: gateway.networking.k8s.io/v1

kind: Gateway 

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

apiVersion: gateway.networking.k8s.io/v1

kind: Gateway

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: Gateway
```

```
metadata:
```

```
  name: Gateway
```

```
  namespace: gateway-ns
```

```
  annotations:
```

```
    cert-manager.io/cluster-issuer: clusterissuer ←
```

```
spec:
```

```
  gatewayClassName: envoy
```

```
  listeners:
```

```
  - name: http-listener
```

```
    port: 80
```

```
    protocol: HTTP
```

```
    allowedRoutes:
```

```
      namespaces:
```

```
        from: All
```

apiVersion: gateway.networking.k8s.io/v1

kind: Gateway

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy ←

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

apiVersion: gateway.networking.k8s.io/v1

kind: Gateway

metadata:

name: Gateway

namespace: gateway-ns

annotations:

cert-manager.io/cluster-issuer: clusterissuer

spec:

gatewayClassName: envoy

listeners:

- name: http-listener

port: 80

protocol: HTTP

allowedRoutes:

namespaces:

from: All

- name: traefik-https-listener

port: 443

protocol: HTTPS

hostname: demo.programmingwithwolfgang.com

allowedRoutes:

namespaces:

from: All

tls:

certificateRefs:

- group: ""

kind: Secret

name: demo-secret

```
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```



```
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```



# Routing

# Gateway Routing

Listeners can listen on different ports

Different protocols:

- HTTP
- HTTPS
- TLS
- TCP
- UDP

# Routes

Weighted load balancing (traffic splitting)

Uses standards instead of annotations

Routes:

- HTTPS and HTTP
- TCP and UDP
- GRPC
- TLS

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute ←
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
        backendRefs:
          - name: traefik
            port: 80
```

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
        backendRefs:
          - name: traefik
            port: 80
```

```
apiVersion: gateway.networking.k8s.io/v1
```

```
kind: HTTPRoute
```

```
metadata:
```

```
  name: traefik-http-route
```

```
  namespace: traefik-ns
```

```
spec:
```

```
  parentRefs:
```

```
    - name: gateway
```

```
      namespace: gateway-ns
```

```
  hostnames:
```

```
    - "demo.programmingwithwolfgang.com" ←
```

```
  rules:
```

```
    - matches:
```

```
      backendRefs:
```

```
        - name: traefik
```

```
          port: 80
```

```
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
        backendRefs:
          - name: traefik
            port: 80
```

# Routing

Routing options:

- Path
- Header, e.g. User-Agent
- Query
- HTTP method
- Redirect request



hostnames:

- "demo.programmingwithwolfgang.com"

rules:

- backendRefs:

- name: app-one ←

port: 80

weight: 20 ←

- name: app-two

port: 80

weight: 80

hostnames:

- "demo.programmingwithwolfgang.com"

rules:

- backendRefs:

- name: app-one

port: 80

weight: 20

- name: app-two ←

port: 80

weight: 80 ←

rules:

- matches:
  - path:
    - type: PathPrefix
    - value: /my/path
- filters:
  - type: URLRewrite
    - urlRewrite:
      - path:
        - type: ReplaceFullPath
        - replaceFullPath: /replaced
- backendRefs:
  - name: app-one
    - port: 80
- backendRefs:
  - name: app-two
    - port: 80

rules:

- matches:
  - path:
    - type: PathPrefix
    - value: /my/path

filters:

- type: URLRewrite
  - urlRewrite:
    - path:
      - type: ReplaceFullPath
      - replaceFullPath: /replaced

backendRefs:

- name: app-one
  - port: 80
- backendRefs:
  - name: app-two
    - port: 80

rules:

- matches:
  - path:
    - type: PathPrefix
    - value: /my/path
  - filters:
    - type: URLRewrite
      - urlRewrite:
        - path:
          - type: ReplaceFullPath
          - replaceFullPath: /replaced

backendRefs:

- name: app-one
  - port: 80
- backendRefs:
  - name: app-two
    - port: 80

rules:

- matches:
  - path:
    - type: PathPrefix
    - value: /my/path
  - filters:
    - type: URLRewrite
      - urlRewrite:
        - path:
          - type: ReplaceFullPath
          - replaceFullPath: /replaced
    - backendRefs:
      - name: app-one
        - port: 80
      - backendRefs:
        - name: app-two
          - port: 80

# Gateway API vs Ingress

# Gateway API vs Ingress

Gateway API might not replace Ingress

Ingress is very easy to manage

- Especially in smaller teams

Gateway might be better for multi-team setups due to modularity



# Gateway Controller

# Gateway Controller

Nginx Gateway Fabric

Traefik

Envoy Gateway

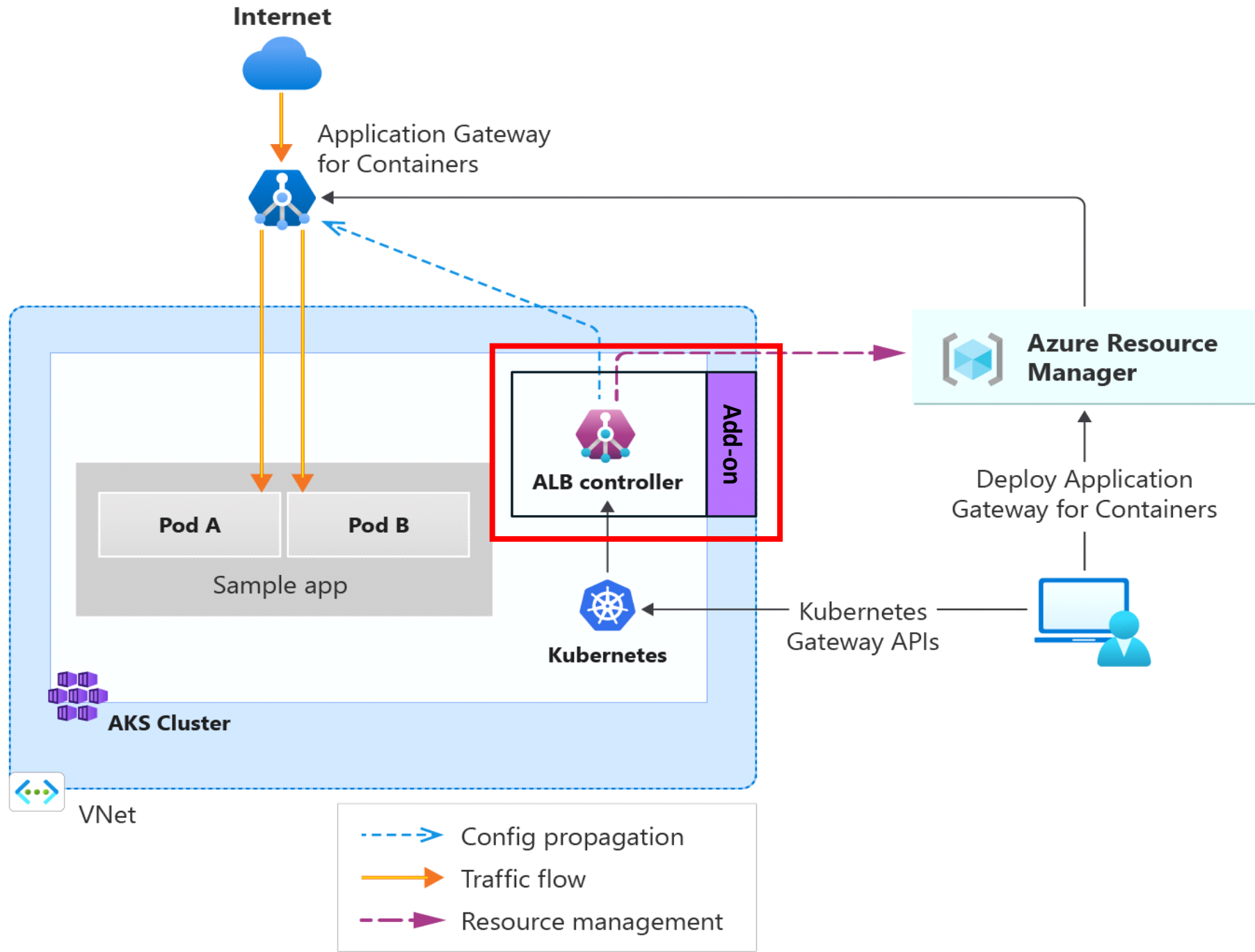
Azure Application Gateway for Containers

# Application Gateway for Containers

AKS addon-on in preview

Brings additional features:

- Advanced health checks
- WAF support
- Monitoring
- Autoscaling and high-availability



Demo

# Thank you for watching!

Remember to leave your questions and rate the presentation in the section below.



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